

Benchmarking Classical Grid-Based Planners in ROS 2

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Course: RBE 550 – Motion Planning

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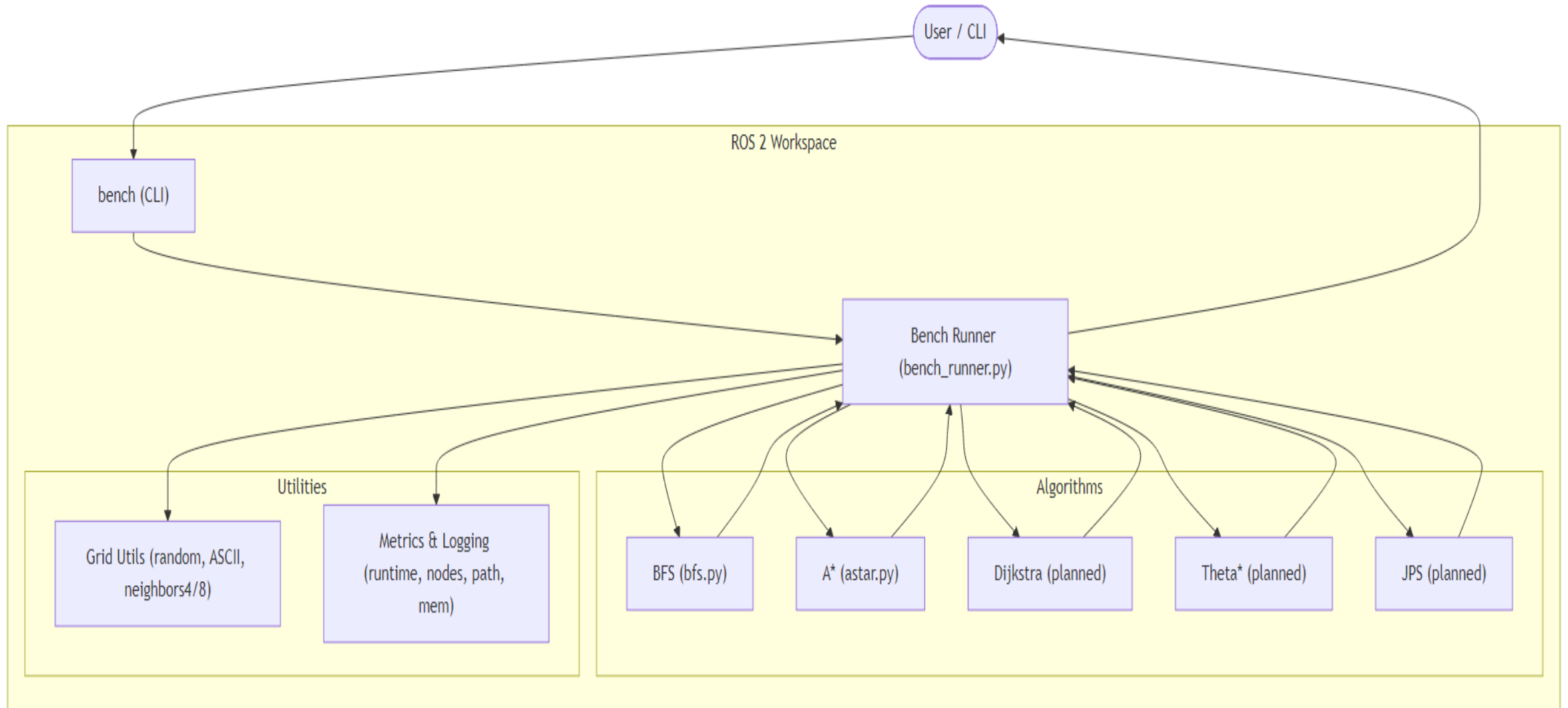
Motivation

- Motion planning algorithms trade off between speed, optimality, and memory.
- Benchmarking these planners fairly is difficult without consistent environments.
- This project creates an automated, reproducible ROS 2 benchmarking framework.

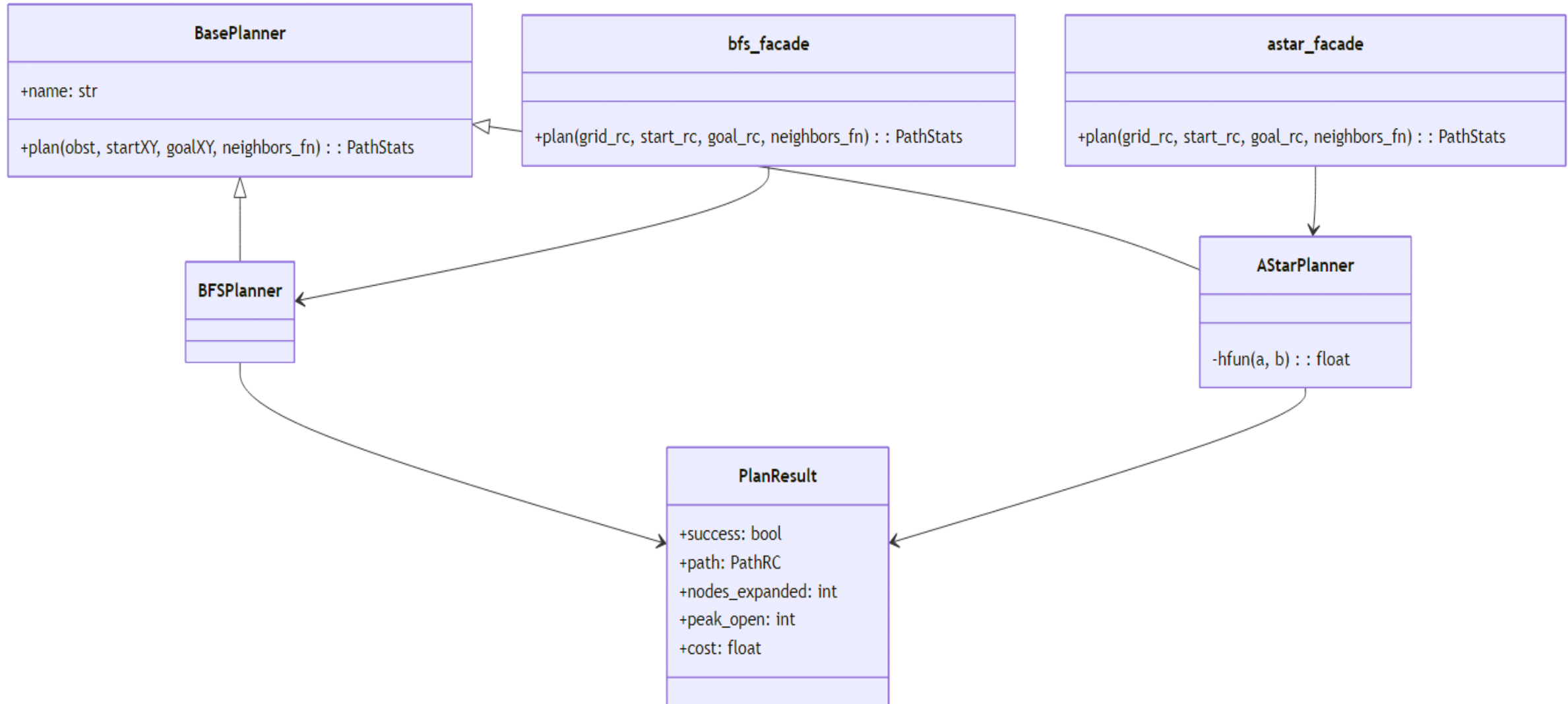
Objectives

- Implement multiple search algorithms:
BFS, Dijkstra, Greedy Best-First, A*.
- Generate both random and structured (maze) environments.
- Record metrics: **path length, runtime, nodes expanded.**
- Package the system using Docker for full reproducibility.

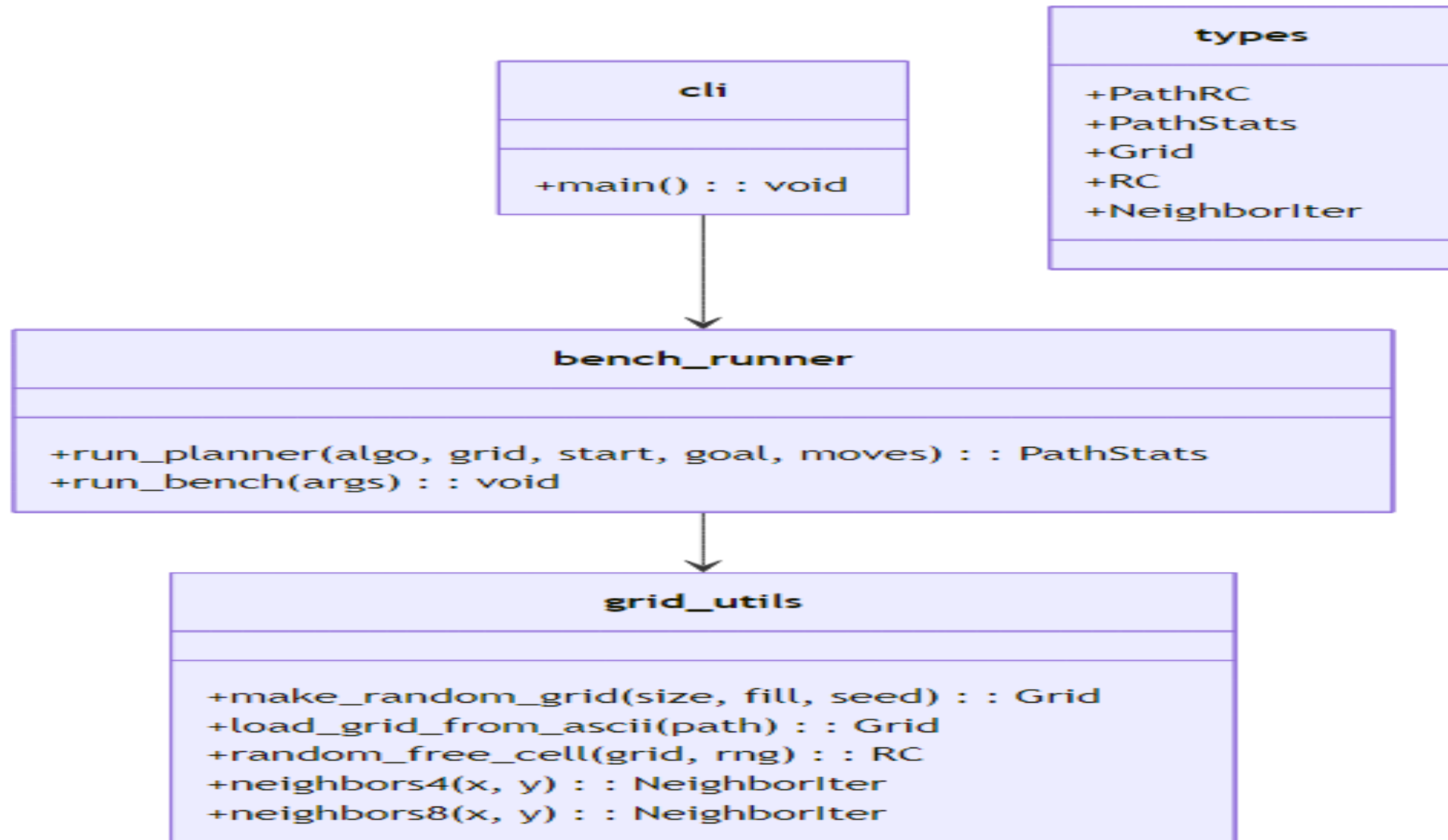
System Architecture



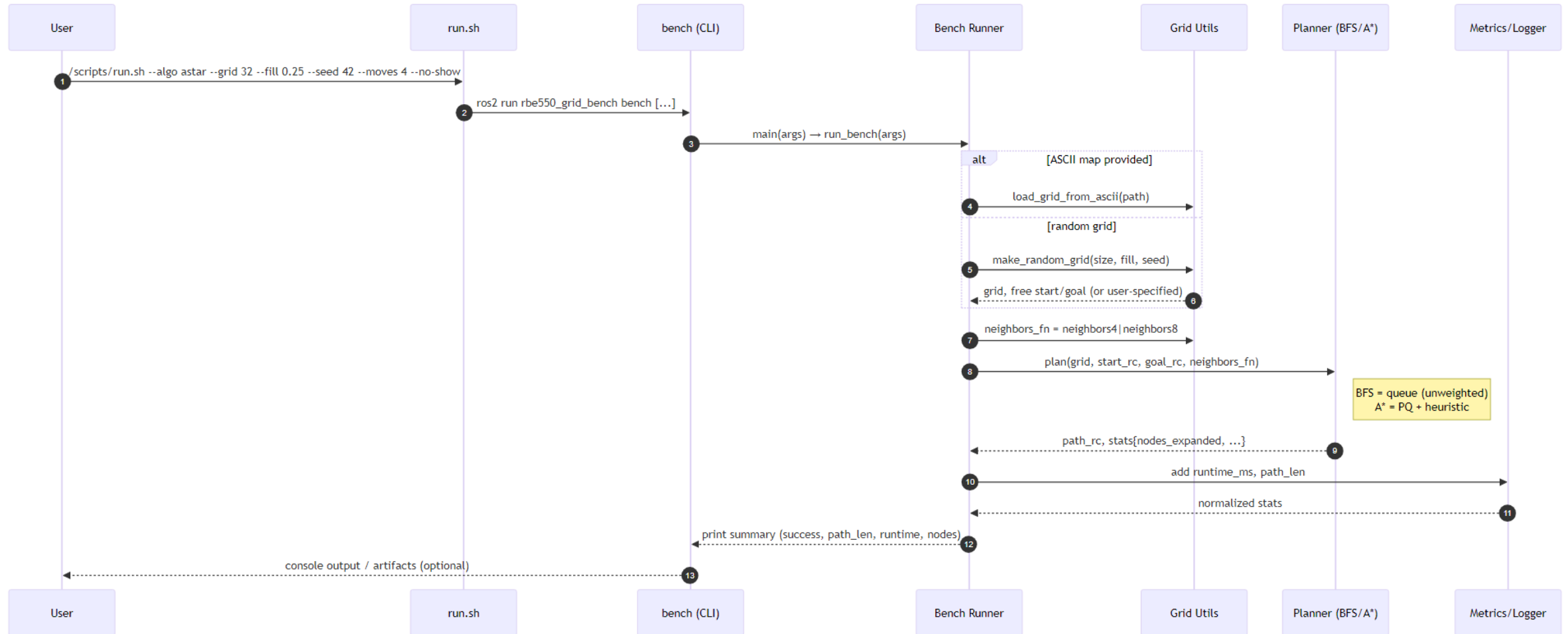
Class Structure (Algorithms)



Class Structure (Core & Environment)



Sequence Diagram



Packaging & Reproducibility

- `git clone git@github.com:meljahmi-personal/RBE550-Workspace.git`
- `cd RBE550-Workspace`
- `./scripts/run_docker.sh --steps 10 --render-every 2 --no-show`
- Builds Docker image `rbe550-bench`
- Runs benchmark automatically
- Saves results to `./outputs/`

Preliminary Results

Algorithm	Connectivity	Heuristic	Path Length	Nodes Expanded	Runtime (ms)
BFS	4	-	64	2155	2.39
A*	4	Manhattan	64	910	1.93
A*	8	Euclidean	45	455	1.81

- A* expands fewer nodes
- Heuristics reduce exploration
- 8-connected + Euclidean shortens path

Validation

- Deterministic seeding for reproducible random maps.
- Maze and random environments tested repeatedly.
- Results are identical across different systems using Docker.

Challenges & Next Steps

- **Challenges:**

- Fixing COLCON_TRACE build issue.
- Ensuring maps/ inclusion inside Docker.
- Balancing parameters for fair comparisons.

- **Next Steps:**

- Add weighted heuristics and cost variants.
- Integrate visualization (RViz / Matplotlib).
- Extend to probabilistic planners later in the semester.

Repositories & References

- Code:** `git@github.com:meljahmi-personal/RBE550-Workspace.git`
- Docs:** `git@github.com:meljahmi-personal/RBE550-project-proposal.git`
- Folder:** `project_status_October_6/status/`
- References:**
 - ROS 2 Navigation documentation
 - WPI RBE550 course materials
 - Standard BFS, Dijkstra, and A* algorithm literature